

When and Why Advice-Taking Strategies Differ:

A Dual-Hurdle-Based Meta-Analysis

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Advice-Taking

Studies on advice taking often use a Judge Advisor System setup

- Judge makes initial judgement
- Judge receives advice
- Judge makes final judgement

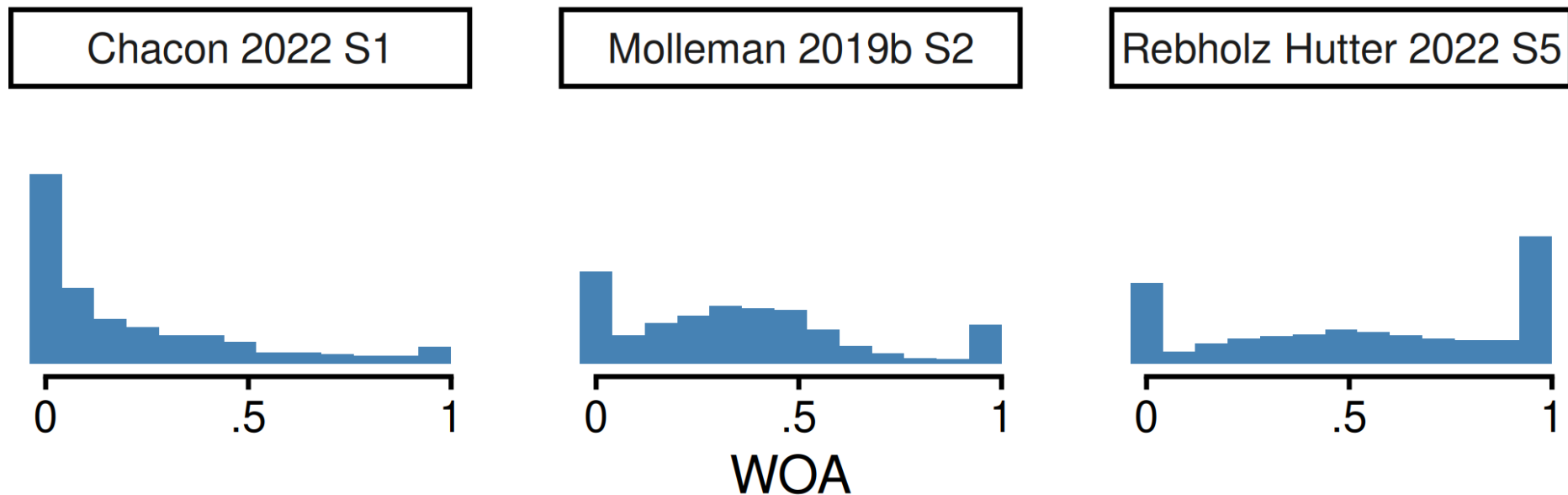
Outcome metric commonly “Weight of Advice” (WOA)

$$WOA = \frac{|\text{initial} - \text{final}|}{|\text{initial} - \text{advice}|}$$

$WOA = 0$	Decline
$WOA = 1$	Adopt
$0 < WOA < 1$	Compromise

Weight of Advice

WOA continuous metric, but often empirically see concentrations of values at 0 and 1



Typically are modeled with standard linear regression procedures

- Assume normally distributed residuals!

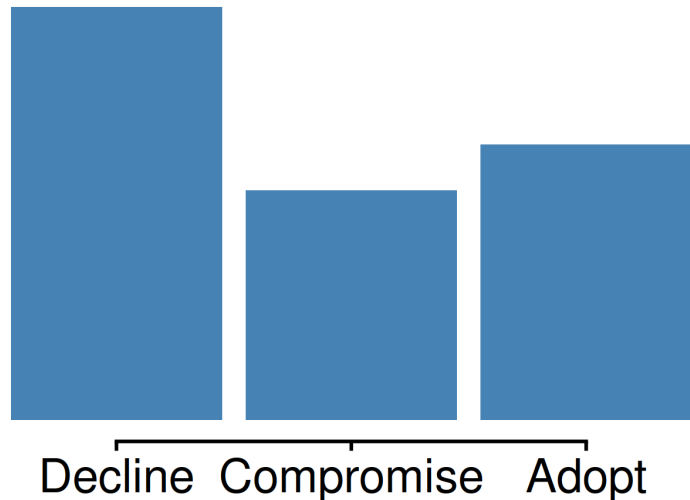
Dual-Hurdle Model

Separates outcome into two phases:

Discrete

Decline, adopt, or compromise?

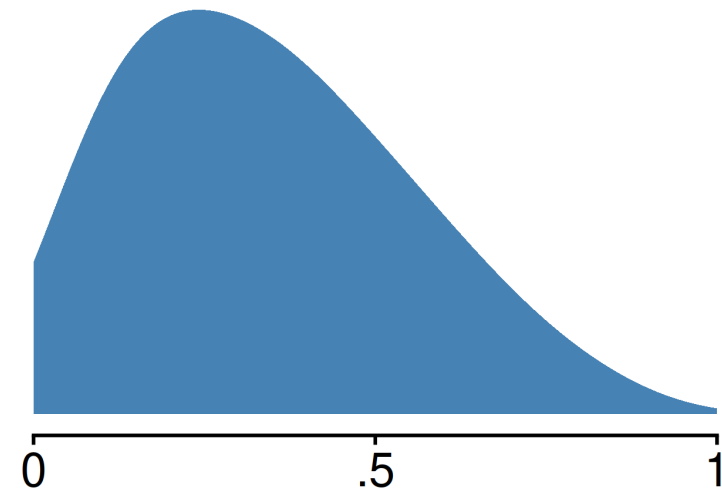
Categorical logit: estimates probability of each choice



Continuous

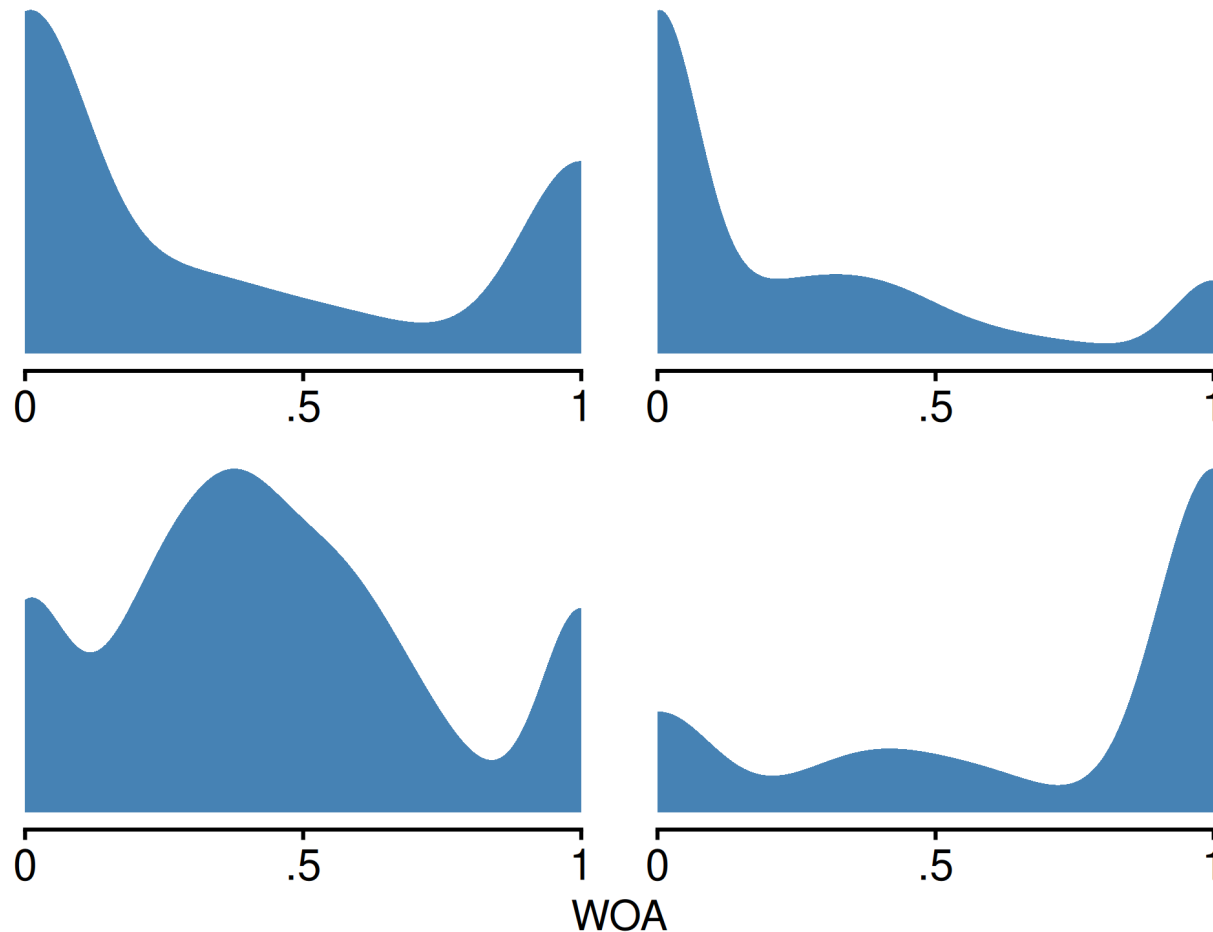
If compromise, how much?

Beta distribution: estimates continuous WOA between 0 and 1



Dual-Hurdle Model

Makes WOA distributions like...



Can examine effects of predictors for both the categorical and continuous pieces

Captures variation at the item- and person-level

Present study: extend dual-hurdle model to meta-analytic context

- Account for study-level variation

Present Study

Used data collected for an advice-taking metaanalysis on gender effects (Kausel et al., 2024)

- Full dataset included 94 studies and 308,230 observations.

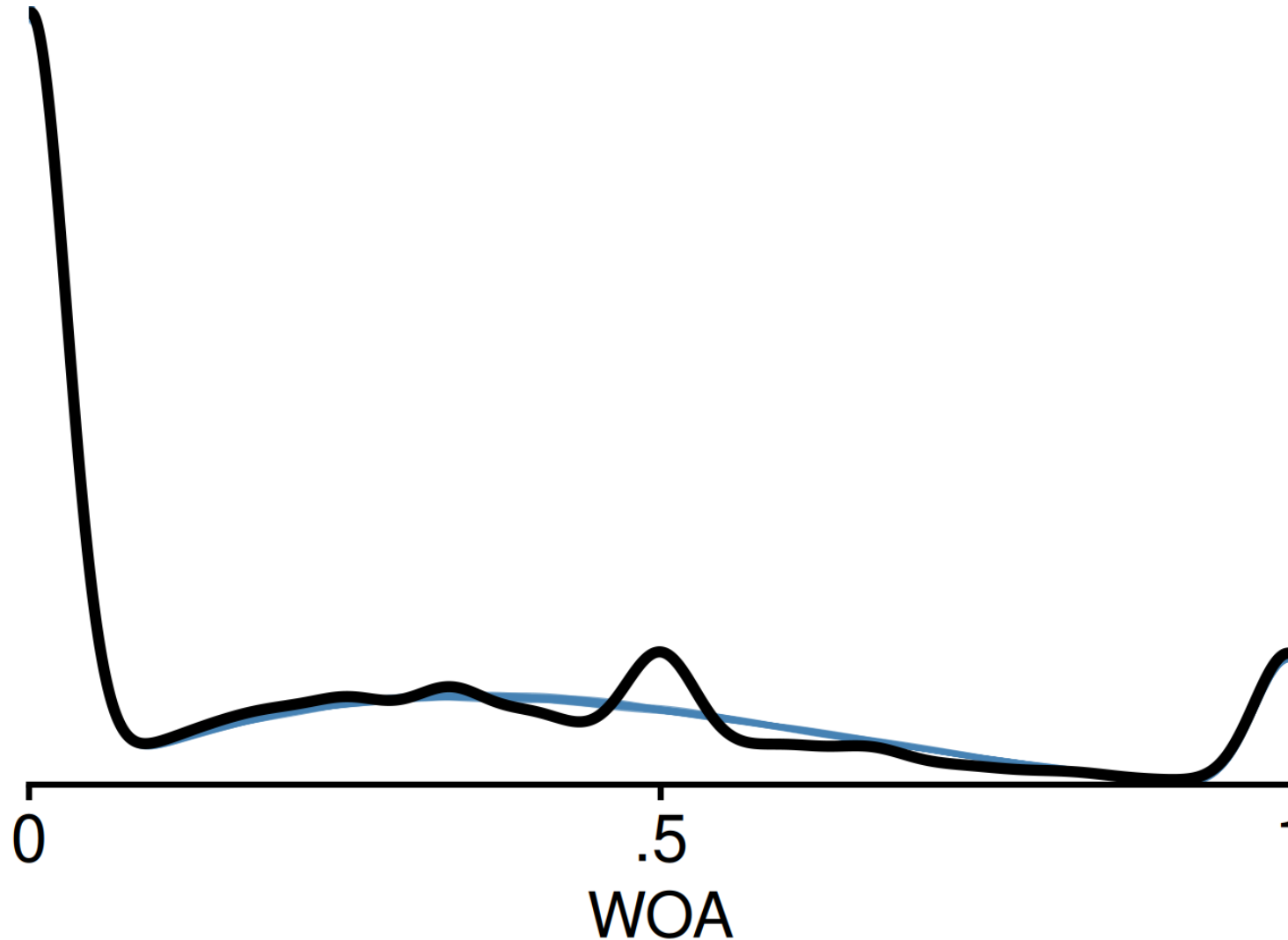
Removed cases that had missing data for WOA or any of the predictors

- Remaining data had 59 studies: 220,134.

Predictors		
Trial-Level	Person-Level	Study-Level
○ Advice Distance	○ Age ○ Gender	○ Almanac Item ○ Future Item ○ Expert Advisor ○ Student Population ○ Incentive

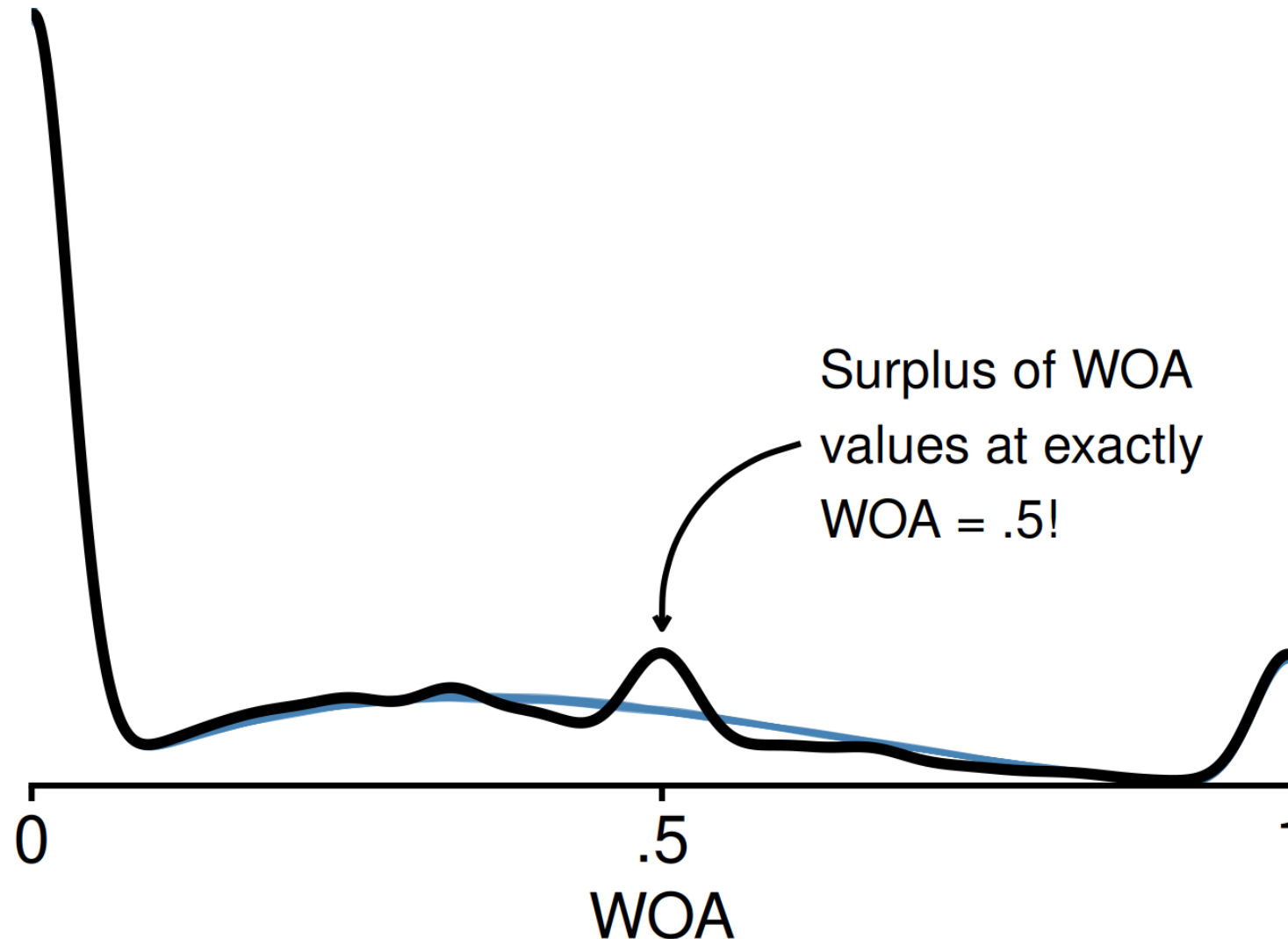
Dual-Hurdle Model

Distribution of WOA values as per the **data** and the **model predictions**



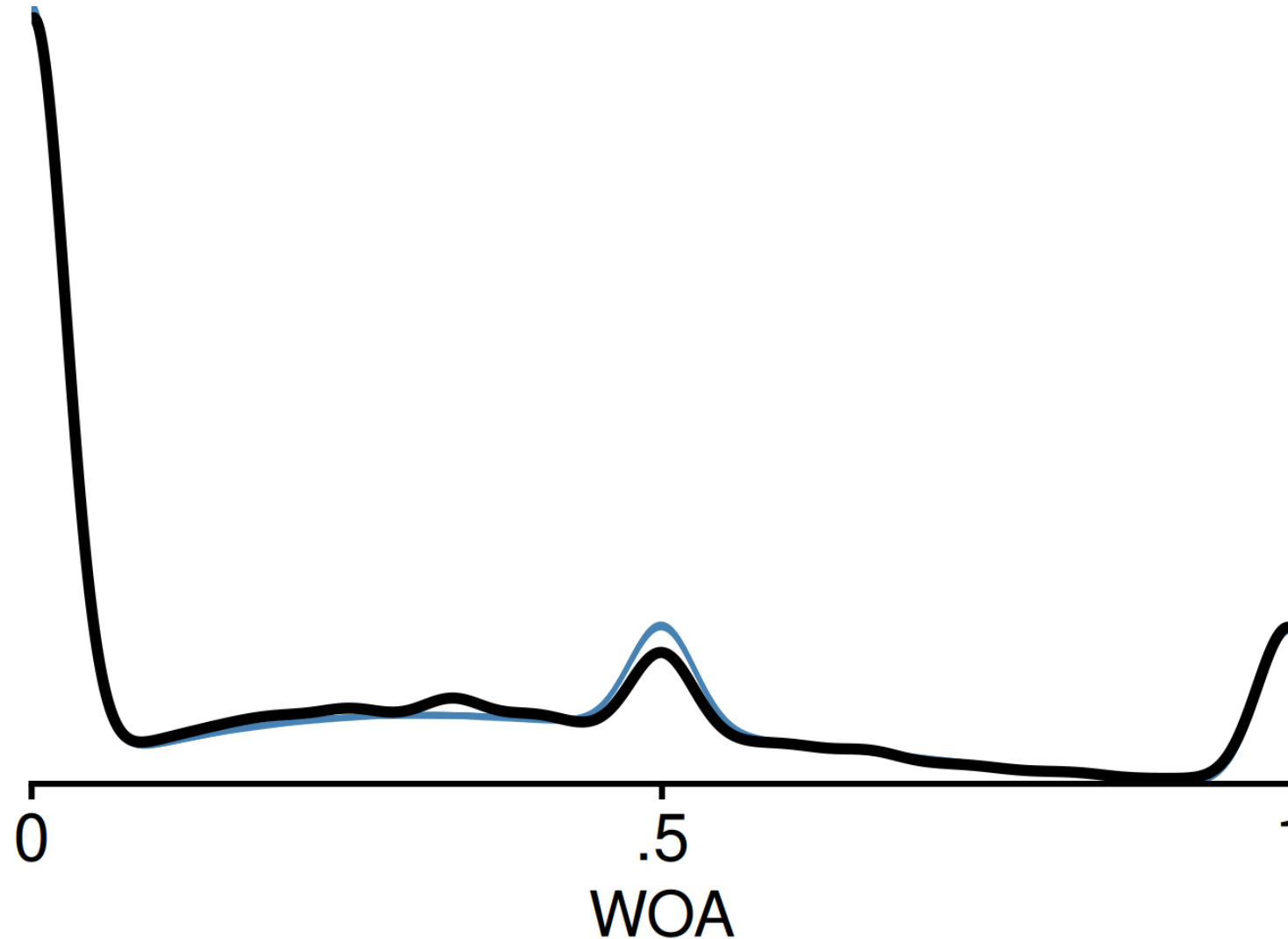
Dual-Hurdle Model

Distribution of WOA values as per the **data** and the **model predictions**

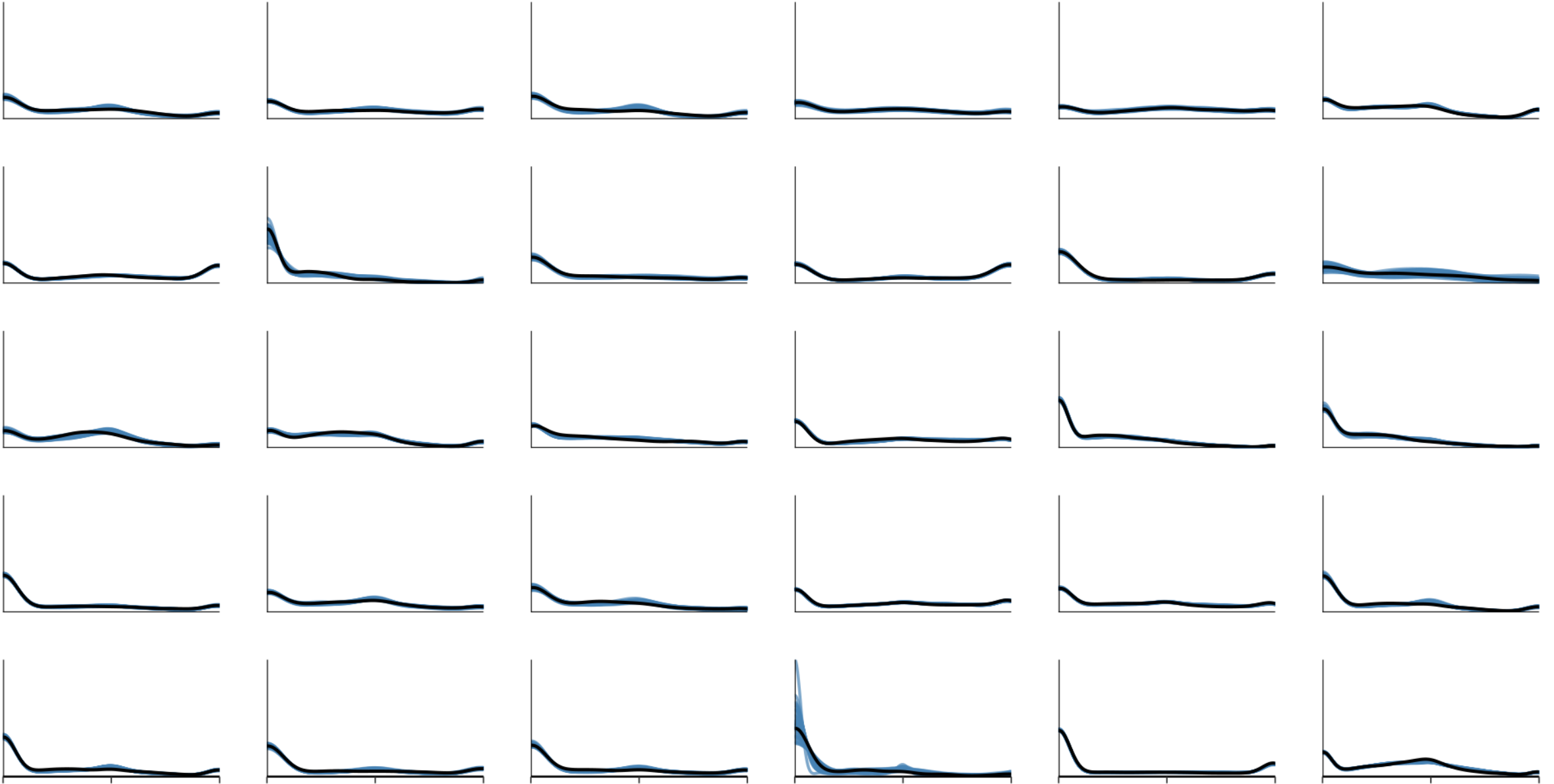


Tri-Hurdle Model

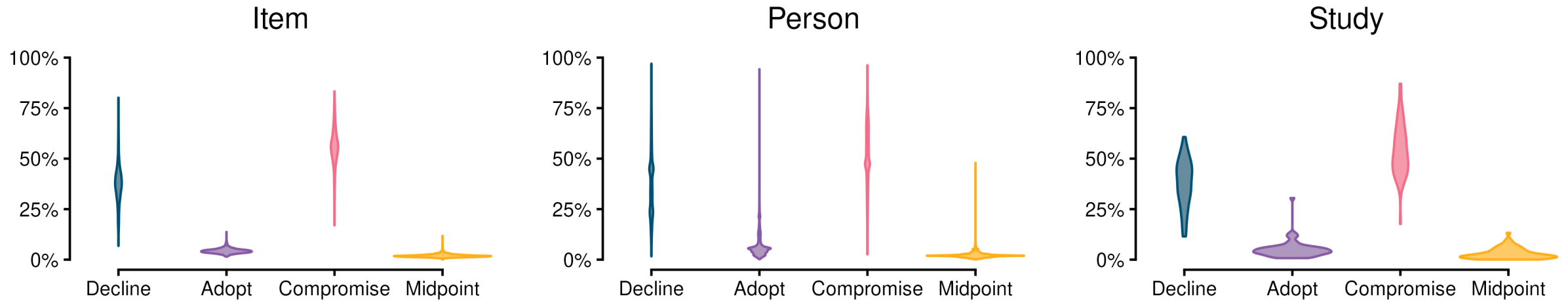
Added a categorical choice: Decline, Adopt, **Midpoint**, Compromise



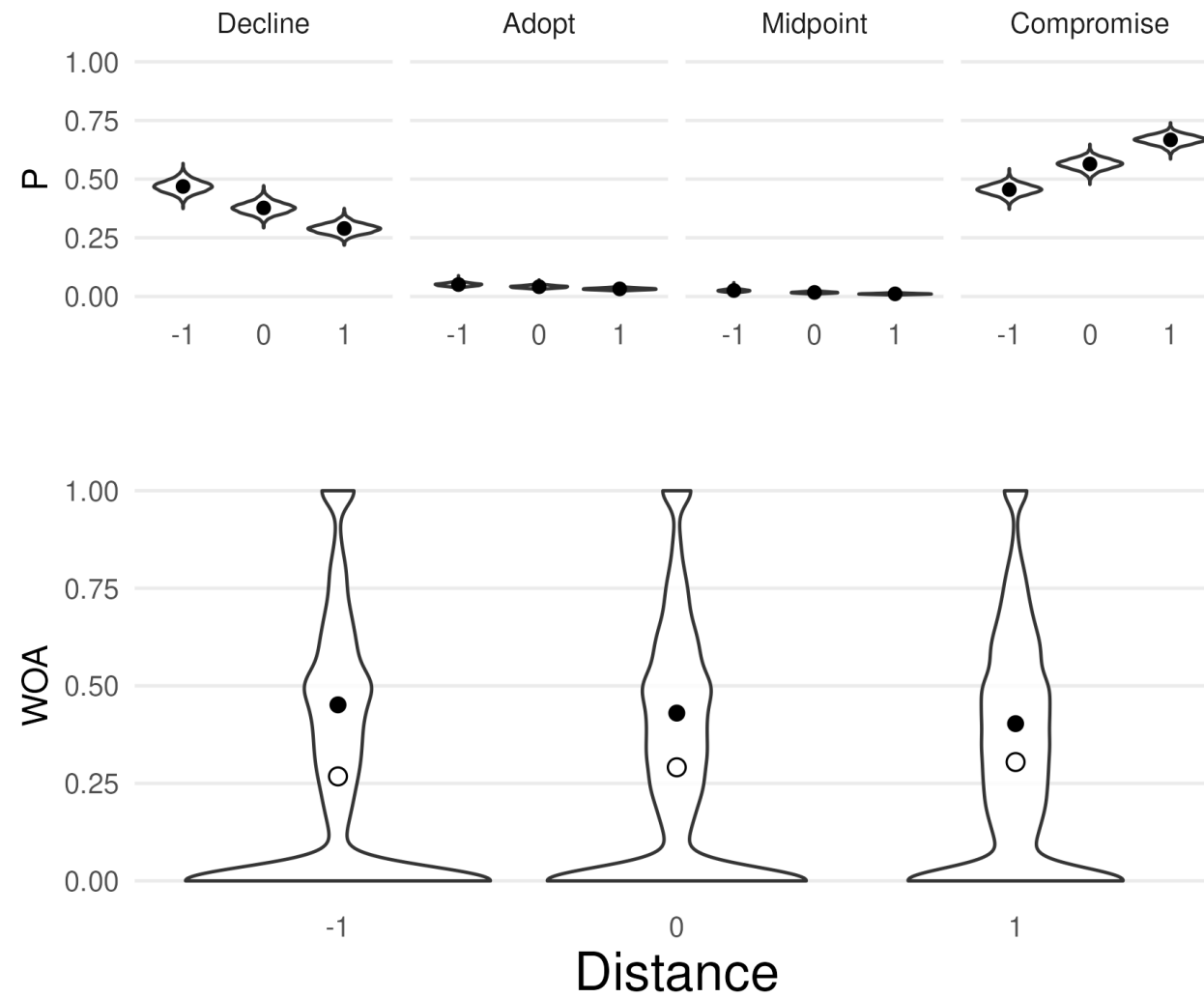
Tri-Hurdle Model



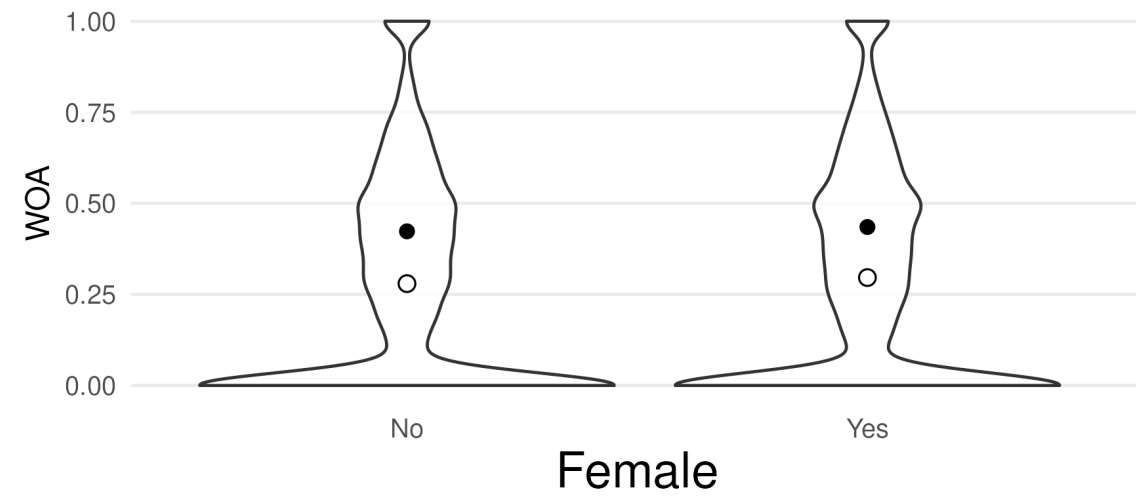
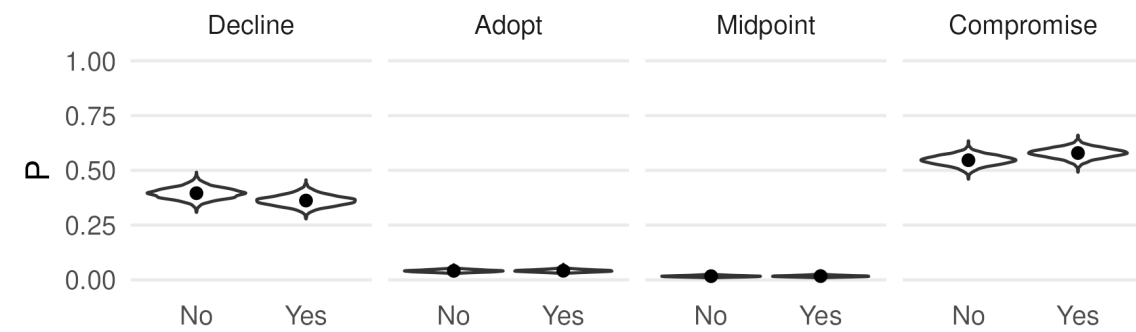
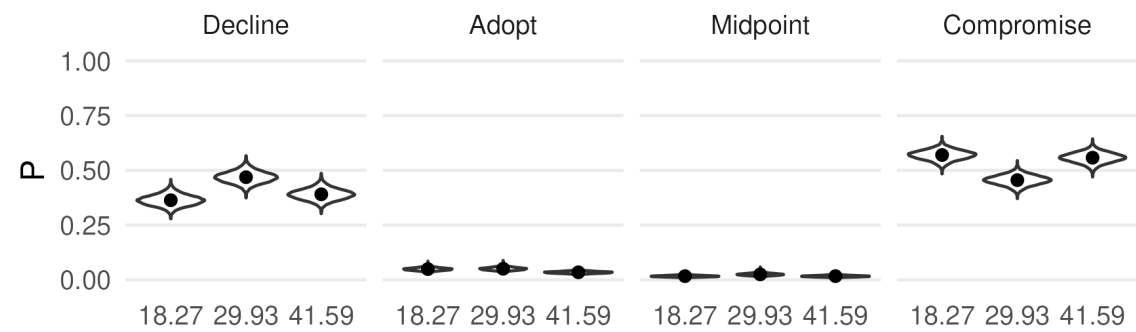
Decline, Adopt, Midpoint, Compromise



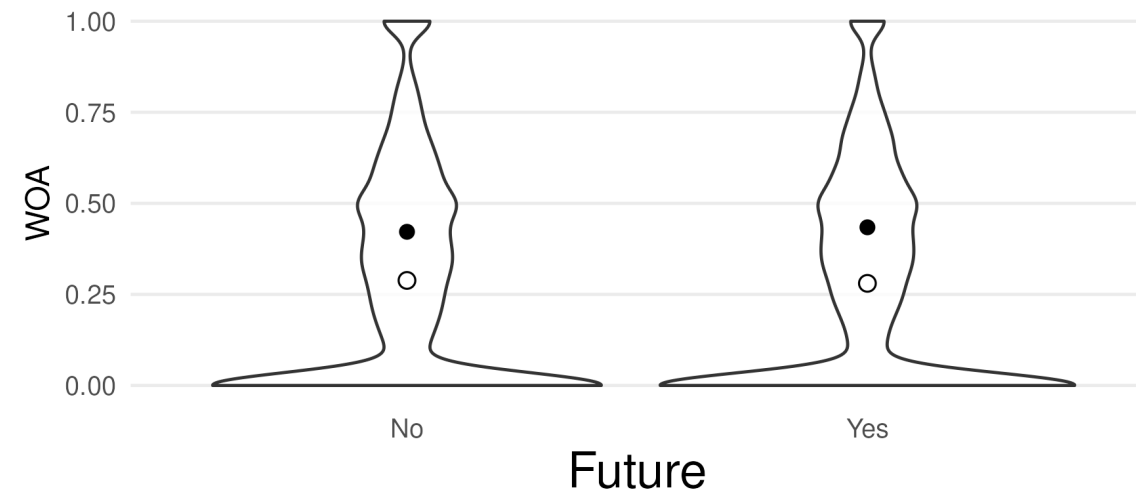
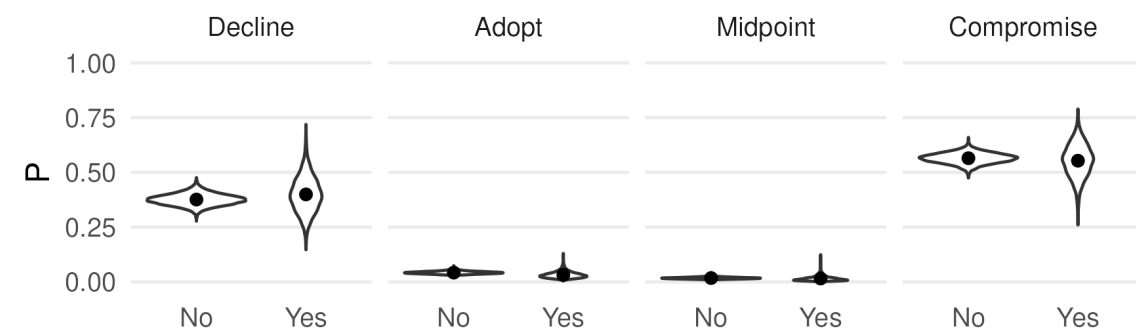
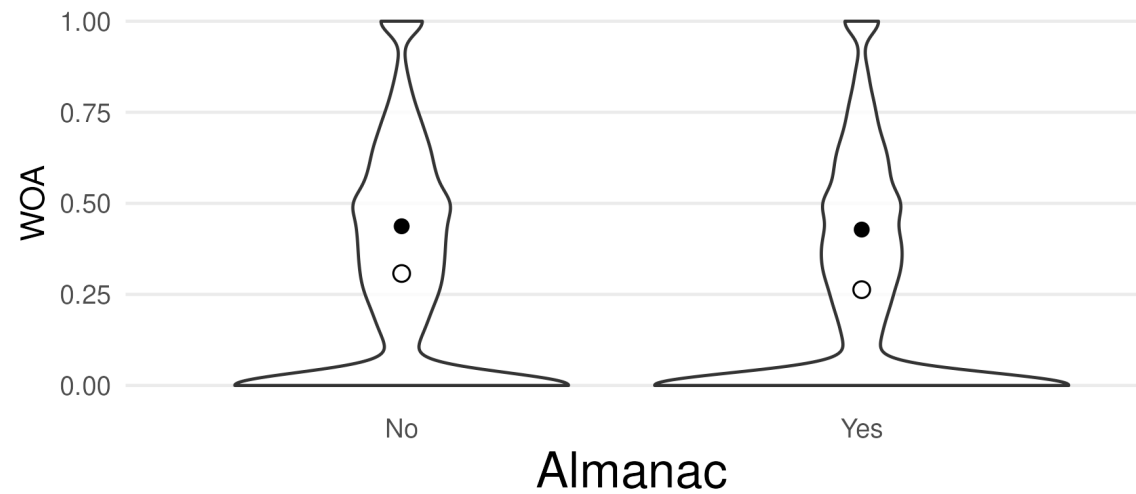
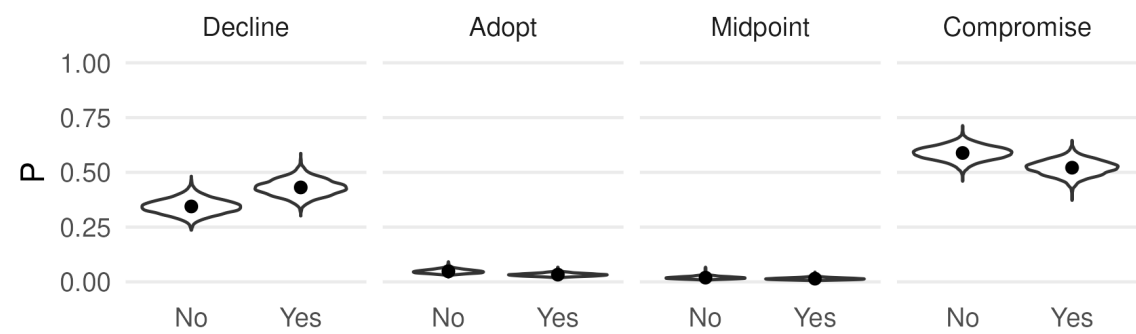
Distance



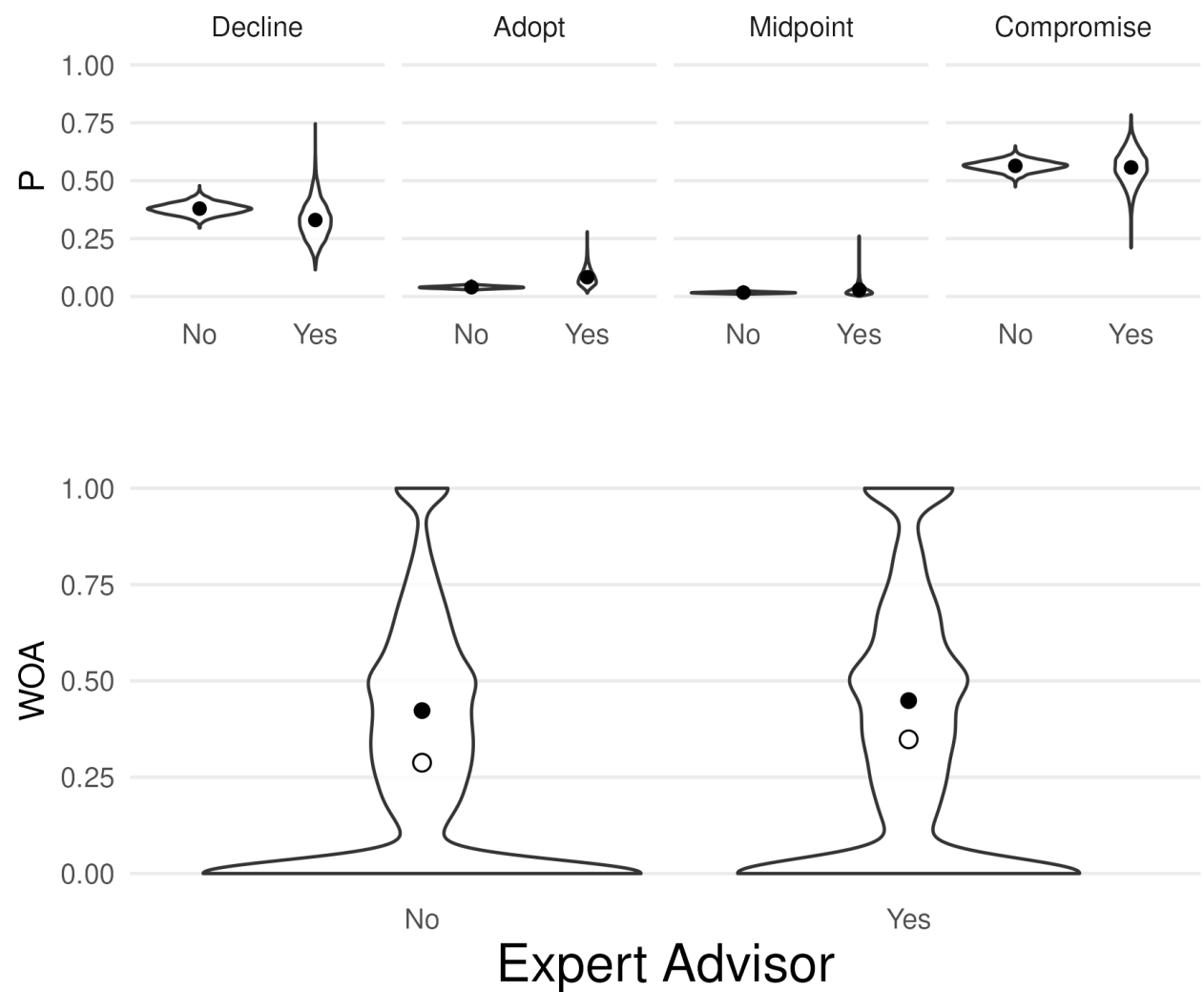
Age and Gender



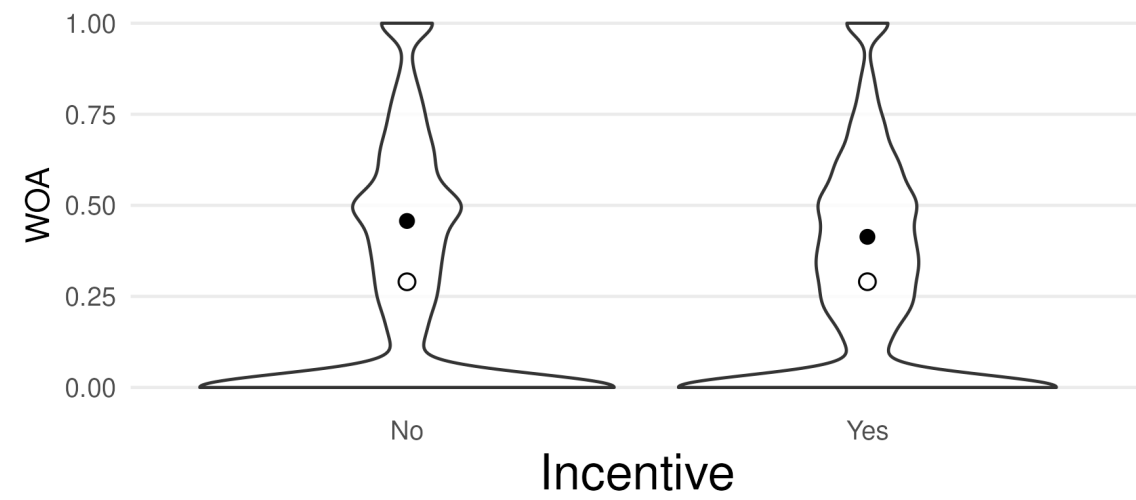
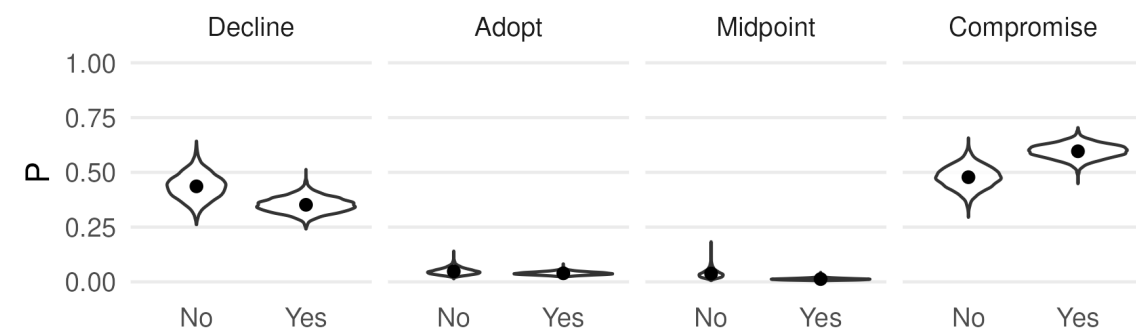
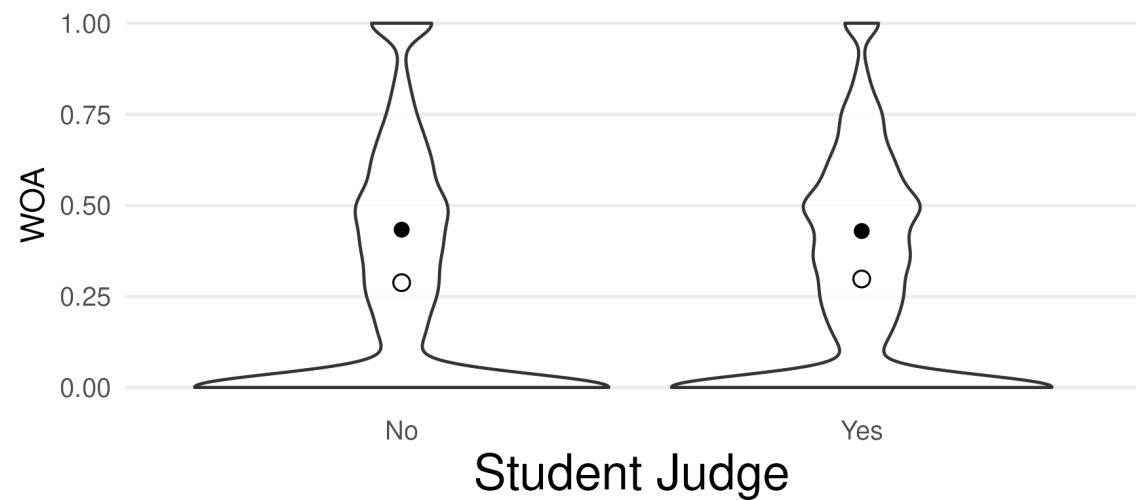
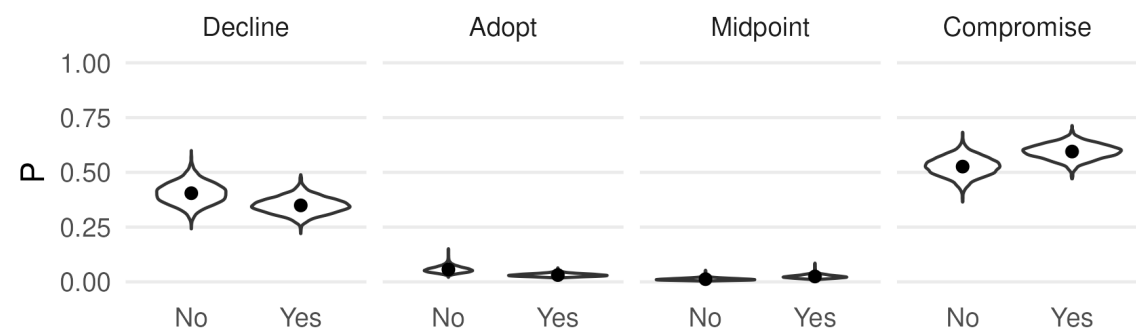
Study-Level Predictors



Study-Level Predictors



Study-Level Predictors



Discussion

Midpoint averaging common, could be relevant to include in model

- Varies by study

Differences between student and non-student populations

- Declining a lot, but updating more when compromise

Differences between incentivised and non-incentivised studies

References

Kausel, E.E., Ventura, S., Reyes, V. (November 2024). [Conference presentation]. Gender Differences in Advice Taking: A Meta-Analytic Path Model. Society of Judgement and Decision Making, New York, NY.

Himmelstein, M. (2022). Decline, adopt or compromise? A dual hurdle model for advice utilization. Journal of Mathematical Psychology, 110, 102695. <https://doi.org/10.1016/j.jmp.2022.102695>

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Thank You!

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